

Chapter 4: Analysis II

Optimization

Julian Klix | **E600 Mathematics**

August 26, 2025



Outline

In this chapter, we discuss

- The formal basics of mathematical optimization
- Unconstrained optimization and its justification
- Optimization with one equality constraint and its justification
- Generalization to more complex problems
- Solution techniques (especially: simplification)

Constrained Optimization – Introduction

- Unconstrained optimization problem

$$\underset{x \in \text{dom}(f)}{\text{maximize}} \quad f(x)$$

- Optimization problem with equality constraints

$$\underset{\text{dom}(f)}{\max} \quad f(x) \quad \text{subject to} \quad g_i(x) = 0, \quad i = 1, \dots, m$$

- General constrained optimization problem

$$\underset{\text{dom}(f)}{\max} \quad f(x) \quad \text{s.t.} \quad \begin{aligned} g_i(x) &= 0, \quad i = 1, \dots, m \\ h_i(x) &\leq 0, \quad i = 1, \dots, k \end{aligned}$$

⇒ We focus on maximization problems → fully analogous for minimization!

Language for Optimization Problems

- **(Global) maximum** for f if $\forall x \in X : f(x_0) \geq f(x)$
 - $\Rightarrow$ maximum refers to function value $f(x_0) = \max f(x)$
 - $\Rightarrow$ maximizer refers to value $x_0 = \arg \max f(x)$
- **Local maximum** if $\exists \varepsilon > 0$ such that $\forall x \in X \cap B_\varepsilon(x_0) : f(x_0) \geq f(x)$
 - $\Rightarrow$ Strict versions: inequalities strict for all $x \neq x_0$
- Global maximization implies local maximization
 - $\Rightarrow$ Definitions analogously for minima and minimizers
- **Extremum** (global or local): Either maximum or minimum
- **Critical point**: potential candidate for an extremum (e.g. FOC)

Definitions for Optimization Problems

- **Maximizer** (alternatively maximum argument):

$$\arg \max_{C(\mathcal{P})} := \{x^* \in C(\mathcal{P}) : f(x^*) = \max_{C(\mathcal{P})} f(x)\}$$

⇒ Generally a set $\rightarrow$ treated as the element for singletons ($\Leftrightarrow$ uniqueness)!

- **Constraint set** of a problem $\mathcal{P}$: subset in domain of f

$$C(\mathcal{P}) := \{x \in X : \forall i \in \{1, \dots, m\} \wedge \forall j \in \{1, \dots, k\} : g_i(x) = 0 \wedge h_j(x) \leq 0\}$$

- **Restricted function** for a subset $A \subseteq \text{dom}(f)$: $f|_A : A \mapsto \mathbb{R}, x \mapsto f(x)$
- **Constrained maximizer** of f in the problem $\mathcal{P}$: maximizer of $f|_{C(\mathcal{P})}$

Extremum Existence

Weierstrass Extreme Value Theorem

Suppose that $X \subseteq \mathbb{R}^n$ is **compact**, and that $f: X \mapsto \mathbb{R}$ is **continuous**. Then, f assumes its maximum and minimum on X , such that

$$\arg \max_{x \in X} f(x) \neq \emptyset \quad \text{and} \quad \arg \min_{x \in X} f(x) \neq \emptyset$$

- Includes multivariate (real-valued) functions
- Recall: In Euclidean vector spaces $\mathbb{X} = (\mathbb{R}^n, +, \cdot)$ compact $\Leftrightarrow$ closed & bounded
 - Closedness: Extremum as corner solution is included
 - Boundedness: Extremum is reached (e.g. monotone functions)
 - Continuity: Extremum isn't part of a jump

Roadmap for Unconstrained Optimization

- General maximization approach
 - ⇒ Reduce domain to set of candidate points
 - ⇒ Find all local maximizers and see which are global (comparing values)
- Outline
 - “Good” necessary conditions give a relatively narrow set of candidates
 - ⇒ Generally, first order conditions (FOC) are used
 - Further candidates: boundary points, points of non-differentiability
 - Sufficient conditions further restrict the set of candidates
 - Existence is useful to guarantee that at least one candidate is a solution

Necessary Conditions for Local Maximizers: FOC

Critical Point or Stationary Point

Let $X \subseteq \mathbb{R}^n$, $f: X \mapsto \mathbb{R}$ and $x^* \in X$. Then, if f is differentiable at x^* and $\nabla f(x^*) = \mathbf{0}$, we call x^* a critical point of f or a stationary point of f .

- **Necessary** FOC: all *interior* local maxima are critical points
- Not sufficient: more points feature $\nabla f(x) = 0$
 $\Rightarrow$ Local minima and "saddle points"
- More insight from second derivative?
 - $f \in C^2(\mathbb{R})$: f' positive before and negative after local maximizer x^*
 $\Rightarrow f'$ decreasing around x^* : $f''(x^*) < 0$
 - Recall: definiteness $\approx$ "sign" of symmetric matrix

Sufficient Conditions for Local Maximizers: SOC

Unconstrained Local Maximum – Sufficient Condition

Let $X \subseteq \mathbb{R}^n$, $f \in C^2(X)$ and $x^* \in \text{int}(X)$. Suppose that x^* is a critical point of f , and that $H_f(x^*)$ is negative definite. Then, x^* is a strict local maximizer of f .

- For univariate functions: $H_f(x^*)$ negative definite $\Leftrightarrow f''(x) < 0$
- Minimum: FOC + Hessian *positive definiteness*
- Careful: what about x^* where
 1. the FOC holds: $\nabla f(x^*) = 0$,
 2. $H_f(x^*)$ is negative semi-definite, but
 3. $H_f(x^*)$ is not negative definite? $\Rightarrow$ Cannot rule out as a solution, compare values to other candidates!

A Shortcut for Definiteness

- Define **leading principal minor** D_k

$$D_k = \det(A_k) \text{ where } A_k = (a_{ij})_{i \in \{1, \dots, k\}, j \in \{1, \dots, k\}}$$

$\Rightarrow$ Submatrix consisting of first k rows and columns

Definiteness and Leading Principal Minors

Let $A \in \mathbb{R}^{n \times n}$ be a symmetrical matrix. Then the following is true

1. A is positive definite if and only if $D_k > 0$ for all k
 2. A is negative definite if and only if $(-1)^k D_k > 0$ for all k
- More general version for semi-definiteness exists (but less useful in practice)

Global Maximization – Helpful Theorems

Sufficiency for the Global Unconstrained Maximum

Let $X \subseteq \mathbb{R}^n$ be a **convex** set, and $f \in C^2(X)$. Then, if f is **concave** and for $x^* \in \text{int}(X)$, it holds that $\nabla f(x^*) = \mathbf{0}$, then x^* is a **global** maximizer of f .

- Does not require compactness
- Quasi-concavity not sufficient $\rightarrow$ saddle points
- For global **minimizers**, f has to be **convex** instead

Maximum and Quasi-Convexity

Let $X \subseteq \mathbb{R}^n$ be a **convex** set, and $f \in C^2(X)$. Then if f is quasi-concave and x_0 a local maximizer, then x_0 is also a global maximizer of f .

A Cookbook Recipe for the Global Maximum

- 1. Determine whether a solution exists at all (optional)
0. Collect border candidates: boundary points, non-differentiability, limits
1. Interior solutions: Finding and eliminating candidates
 - If existence guaranteed and at any step, only one candidate (including border) remains, stop, you found the maximum!
 - Necessary FOC: Initial set of candidates = critical values
 - Necessary SOC: rule out those that violate it
 - If only one candidate (including border) remains
⇒ Existence guaranteed? Or: sufficient condition holds?
2. Multiple candidates remaining: compare values, check existence if not already done

Optimization Problems with Equality Constraints

A Simple Special Case

- Example with single constraint
- Re-write the constrained problem to use our approach to unconstrained problems

$$\max_{\mathbf{x} \in \mathbb{R}^2} u(\mathbf{x}_1, \mathbf{x}_2) \text{ s.t. } y = p_1 \mathbf{x}_1 + p_2 \mathbf{x}_2 \Leftrightarrow \max_{\mathbf{x}_1 \in \mathbb{R}} u\left(\mathbf{x}_1, \frac{y - p_1 \mathbf{x}_1}{p_2}\right)$$

...having solved the constraint for $\mathbf{x}_2 = \frac{y - p_1 \mathbf{x}_1}{p_2}$

- Works if we can find an expression of the constraint for $\mathbf{x}_1$ that is...
 - explicit: we can write down the equation for $\mathbf{x}_1$ in terms of $\mathbf{x}_{-1}$
...vs. **implicit**: we know that there is some function $\mathbf{x}_1 = h(\mathbf{x}_{-1})$
 - global: the function applies to the whole domain
...vs. **local**: applies around a local maximizer
- Particularly promising for $m < n$ linear constraints (see example)

The Lagrangian – A General Approach

Lagrange's Necessary First Order Condition [Single Constraint]

Consider the constrained problem $\max f|_{C(g)}$ where $X \subseteq \mathbb{R}^n$, $f, g \in C^1(X)$ and x^* such that $g(x^*) = 0$. Suppose that $\nabla g(x^*) \neq \mathbf{0}$. Then, x^* is a local maximizer of the constrained problem only if there exists $\lambda \in \mathbb{R} : \nabla f(x^*) = \lambda \nabla g(x^*)$.

If such $\lambda \in \mathbb{R}$ exists, we call it the **Lagrange multiplier** associated with x^* .

- Here: only constraint $g : \mathbb{R}^n \mapsto \mathbb{R}$ instead of multiple constraints
- **Lagrangian function** (or: "Lagrangian"): $\mathcal{L}(\lambda, x) = f(x) - \lambda g(x)$
- Note, $x^* \in X$ satisfies the FOC iff for a $\lambda \in \mathbb{R}$, (λ, x) is a critical value of $\mathcal{L}(\lambda, x)$!
 - $\Rightarrow$ FOC for $x : \nabla f(x^*) - \lambda \nabla g(x^*) = \mathbf{0} \iff \nabla f(x^*) = \lambda \nabla g(x^*)$ [Lagrangian FOC]
 - $\Rightarrow$ FOC for $\lambda : g(x^*) = 0$ [Constraint is satisfied]

The Lagrangian – An Example

Example 1 Find the vector with minimum Euclidean length $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$ in the $\mathbb{R}^2$ that satisfies $x_1 + x_2 = 1$, i.e. solve

$$\min_{x \in \mathbb{R}^2} \|x\|_2 \quad \text{subject to } x_1 + x_2 = 1 \iff \max_{\mathbb{R}^2} -\|x\|_2 \quad \text{s.t. } x_1 + x_2 = 1$$

Alternative (Simple) Solution Rewrite constraint as $x_2 = 1 - x_1$ and solve

$$\min_{x_1 \in \mathbb{R}} x_1^2 + (1 - x_1)^2$$

1. Finding critical values via FOC: $4x_1 - 2 = 0 \iff x_1 = \frac{1}{2} \Rightarrow x_2 = \frac{1}{2}$
2. Checking SOC: $4 > 0 \Rightarrow x^* = (\frac{1}{2}, \frac{1}{2})$ (global) minimizer

The Lagrangian FOC for multiple constraints

Lagrange's Necessary First Order Condition [Multiple Constraints]

Consider the constrained problem $\max f|_{C(g)}$ where $X \subseteq \mathbb{R}^n$ and $f \in C^1(X)$, $g \in C^1(X, \mathbb{R}^m)$. Let x^* be such that $g(x^*) = \mathbf{0}$ and suppose that $\text{rk}(J_g(x^*)) = m$. Then, x^* is a local maximizer of the constrained problem only if there exists $\Lambda = (\lambda_1, \dots, \lambda_m)' \in \mathbb{R}^m : \nabla f(x^*) = \Lambda' J_g(x^*)$.
If such $\Lambda \in \mathbb{R}^m$ exists, we call λ_i the Lagrange multiplier for the i -th constraint.

- Direct generalization of the single-constraint case
- **Multi-constraints Lagrangian function:** $\mathcal{L}(\Lambda, x) = f(x) - \Lambda' g(x)$

$\Rightarrow$ Again, find critical values by checking FOC of $\mathcal{L}$

Optimization Problems with Inequality Constraints

The General Constrained Optimization Problem

- Optimization problem with equality and inequality constraints

$$\begin{aligned} \max_{\text{dom}(f)} f(x) \quad \text{s.t.} \quad & g(x) = \mathbf{0}, \quad g: \mathbb{R}^n \mapsto \mathbb{R}^m \\ & h(x) \leq \mathbf{0}, \quad h: \mathbb{R}^n \mapsto \mathbb{R}^k \end{aligned}$$

- Lagrange problem except for additional **inequality constraints**
- Inequality constraints can be either binding or slack
 - ⇒ Binding constraint $\approx$ equality constraint (with $\mu_j \approx$ Lagrange multiplier)
 - ⇒ Slack constraint: No restriction on optimization (no value cost $\mu_j = 0$)
- **Complementary slackness**: either $h_j(x) = 0, \mu_j \neq 0$ **or** $h_j(x) < 0, \mu_j = 0$
- **Solve individual Lagrangian case-by-case for each slackness-combination!**

Karush-Kuhn-Tucker – A General Approach

Karush-Kuhn-Tucker Theorem

Consider the constrained problem $\max f|_{C(g,h)}$ where $X \subseteq \mathbb{R}^n$ and $f \in C^1(X)$, $g \in C^1(X, \mathbb{R}^m)$, $h \in C^1(X, \mathbb{R}^k)$. Then, if x^* is a local maximum of the constrained problem for which the set $\{\nabla h_j(x^*) : h_j(x^*) = 0\} \cup \{\nabla g_i(x^*) : i \in \{1, \dots, m\}\}$ is linearly independent, there exist $\Lambda^* \in \mathbb{R}^m$ and $\mu^* \in \mathbb{R}^k$ such that (x^*, Λ^*, μ^*) satisfy the optimality conditions:

1. (Feasibility) $\forall j \in \{1, \dots, k\} : h_j(x) \leq 0$ and $\forall i \in \{1, \dots, m\} : g_i(x) = 0$,
2. (FOC for x) $\nabla f(x) = \sum_{i=1}^m \lambda_i \nabla g_i(x) + \sum_{j=1}^k \mu_j \nabla h_j(x)$,
3. (Complementary Slackness) $\forall j \in \{1, \dots, k\} : \mu_j h_j(x) = 0$.

- Linear independence condition fulfils similar role as rank condition before
- Karush-Kuhn-Tucker Lagrangian function: $\mathcal{L}(\lambda, x) = f(x) - \Lambda' g(x) - \mu' h(x)$

Karush-Kuhn-Tucker – An Example

Example 2 Find the vector with maximum Euclidean length $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$ in the $\mathbb{R}_+^2$ that satisfies $x_1 + x_2 = 1$, i.e. solve

$$\begin{aligned} \max_{x \in \mathbb{R}^2} \|x\|_2 \quad & \text{s.t. } x_1 + x_2 = 1 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}$$

or, put differently (noting that $\sqrt{\cdot}$ is a monotone function)

$$\begin{aligned} \max_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad & \text{s.t. } 1 - x_1 - x_2 = 0 \\ & \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix} \leq \mathbf{0} \end{aligned}$$

Recap Chapter 4

Unconstrained Optimization

- Locally solveable by FOCs & SOC
- Global solution requires further single-point comparisons

Equality-constrained Optimization → Lagrange approach

- Additional constraints that are *always* binding
- Use FOCs of Lagrange-function (internalizing constraints)

Inequality-constrained Optimization → Karush-Kuhn-Tucker approach

- Additional constraints that are *sometimes* binding
- Use FOCs of Lagrange-function of all binding constraints

That's all Folks!

Please take a look at the problem set that we will discuss Thursday morning.

✉ julian.klix@uni-mannheim.de

🏢 L7 3-5, Room 3.43

